

Module 3: Strategic Equilibrium: Foundations & Applications

ECO364-THINKING STRATEGICALLY

Part I: Foundations

RATIONALITY AND COMMON KNOWLEDGE

A Review from Module 2.

- ▶ In Module 2, we talked about dominant strategies.
 - ▶ When a player has a dominant strategy, that is the best strategy that she could be playing against anything that her rival is doing.
 - ▶ To check for a dominant strategy, a player only needs to know her own payoffs and how the strategies of the other player(s) affect her payoffs.
 - ▶ *If a player has a dominant strategy, she doesn't need to know the payoffs of the other player(s)!*
- ▶ In Module 2, we also saw that when a game doesn't have dominant strategies for the players, we could check for dominated strategies and eliminate them.
- ▶ After elimination, other strategies might become dominated and can also be eliminated.
- ▶ We carried on with this procedure and called it *successive elimination of dominated strategies*.
- ▶ However, last time, I mentioned that in order for this procedure to work, we need at each step “deeper levels of strategic thinking”.

Rationality and Common Knowledge

- ▶ We typically assume that players are **rational**.
 - ▶ This means, they act in their best interest (maximize payoffs).
 - ▶ Remember, payoffs described what matters for a player in the games. This incorporates notions of selfishness, fairness, equity, etc.
 - ▶ Rational also means that players are able to compute and reason about these payoffs.
- ▶ However, we will also make some assumptions on *players' knowledge*.
 - ▶ In order to make valid the procedure of iterated elimination of dominated strategies, and other solution concepts, we will need to assume that:
 - ▶ Players know that the rest of the players are rational (enough for a first step of deletion).
 - ▶ Players know that the rest of the players know that they are rational (enough for a second step of deletion).
 - ▶ Players know that the rest of the players know that the rest of the players know that they are rational (enough for a third step of deletion).
 - ▶ And so on...
- ▶ This chain of reasoning is a valid argument if the players have what is called **common knowledge**.
- ▶ The purpose of this first batch of slides is to introduce you to this concept and why it is important when analyzing strategic situations.

A logic puzzle

- ▶ Versions of this puzzle are so well-known that it is very difficult to come up with the definitive version. A more politically incorrect version, although very well-known, is called the *puzzle of the cheating wives*.
- ▶ The first instance of this problem known to me, was from the book *Calculus* by Michael Spivak, which I describe here:

University B, once boasted 17 tenured professors of mathematics. Tradition prescribed that at their weekly luncheon meeting, faithfully attended by all 17, any members who had discovered an error in their published work should make an announcement of this fact, and promptly resign. Such an announcement had never actually been made, because no professor was aware of any errors in her or his work. This is not to say that no errors existed, however. In fact, over the years, in the work of every member of the department at least one error had been found, by some other member of the department. This error had been mentioned to all other members of the department, but the actual author of the error had been kept ignorant of the fact, to forestall any resignations.

One fateful year, the department was augmented by a visitor from another university, one Prof. X, who had come with hopes of being offered a permanent position at the end of the academic year. Naturally, he was apprised, by various members of the department, of the published errors which had been discovered. When the hoped-for appointment failed to materialize, Prof. X obtained his revenge at the last luncheon of the year. "I have enjoyed my visit here very much", he said, "but I feel that there is one thing that I have to tell you. At least one of you has published an incorrect result, which has been discovered by others in the department." What happened the next year?"

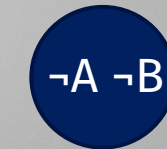
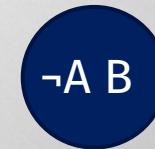
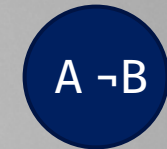
- ▶ Even more intriguing, is that each member of the department already knew what Prof. X asserted, so how could his saying change anything?

How to solve the puzzle?

- ▶ Let's break it into pieces:
 - ▶ Of all the 17 professors, each one of them knows that the others have errors in their work.
 - ▶ Each professor initially is ignorant that there are mistakes in his or her own work.
 - ▶ The rule is that if a professor knows that there is a mistake in his or her own work, a resignation is due.
 - ▶ Then Prof. X comes along, and out of spite, ***publicly announces*** a fact that was already known by each of the professors, namely, that “at least one of you has published an incorrect result”.
 - ▶ Although this fact is known by each of the professors, the public nature of the announcement indeed changed something.
 - ▶ *It altered the level of knowledge of the professors!*
- ▶ I will simplify the problem a little bit to show you exactly what happened and how it can be solved.

Simplifying the puzzle: Only two professors.

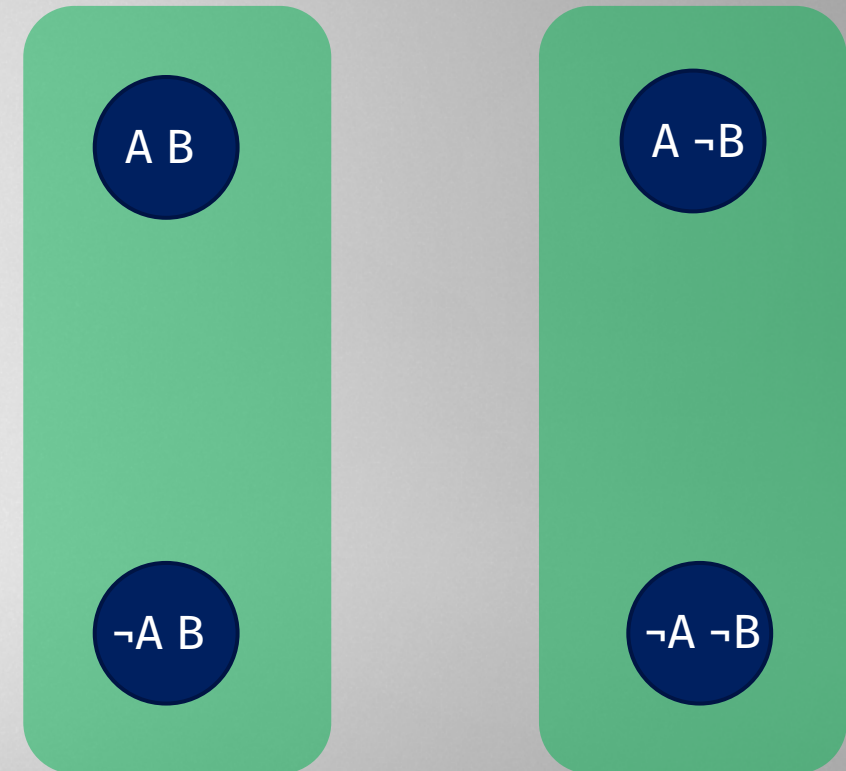
- ▶ To make it simple, consider the case that University B had only two professors in its Mathematics Department: Prof. A and Prof. B.
- ▶ As in the description, Prof A. has an error in his published work, of which Prof. B is aware but Prof. A is not.
- ▶ Prof B. also has an error in her published work, of which Prof. A is aware but Prof. B is not.
- ▶ Let's try to represent the state of knowledge of these two professors.
- ▶ Let A denote the event that Prof. A doesn't have an error in his published work and $\neg A$ the event that Prof A. has an error.
- ▶ Similarly, let B denote the event that Prof. B doesn't have an error and $\neg B$ the event that Prof. B has an error.
- ▶ Thus, there are 4 possible configurations of the state of affairs:



Of course, the actual state is $\neg A \neg B$, since both professors have errors in their published work.

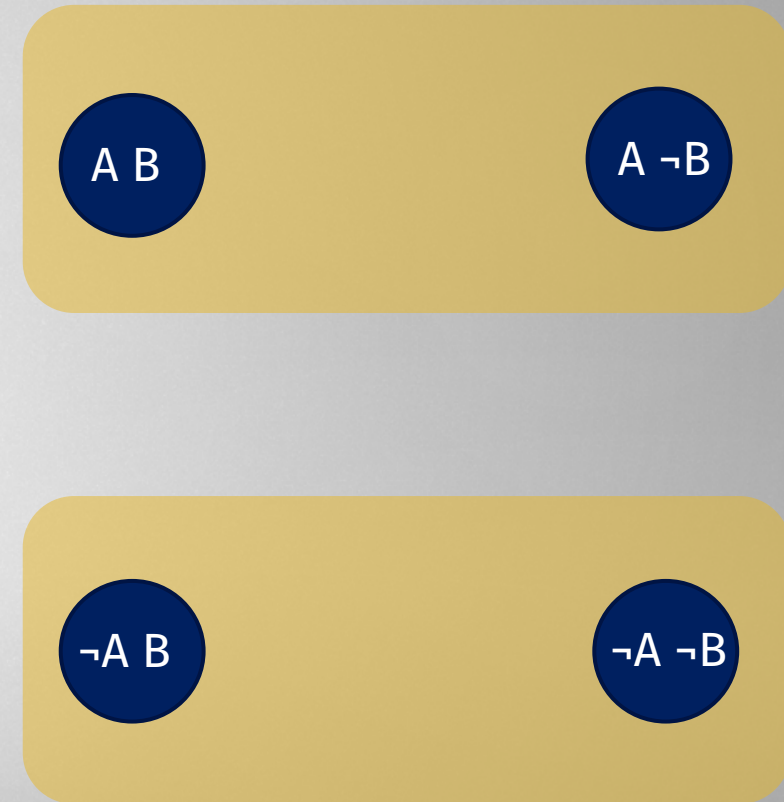
Simplifying the puzzle: Only two professors.

- ▶ Of course, the actual state is $\neg A \neg B$, since both professors have errors in their published work.
- ▶ Let's represent Prof. A's knowledge, in green:
 - ▶ Prof. A knows whether Prof. B has a mistake in her published work or not. Prof. A can distinguish between these two events.
 - ▶ But Prof. A cannot distinguish whether he has a mistake or not in his own work.
 - ▶ The green rectangles represent this, the states which Prof. A can distinguish.
 - ▶ As you can see, the states AB and $\neg AB$ form a group, and the states $A\neg B$ and $\neg A\neg B$ form another.
 - ▶ At the true state, $\neg A\neg B$, Prof. A's information cannot reveal the true state.



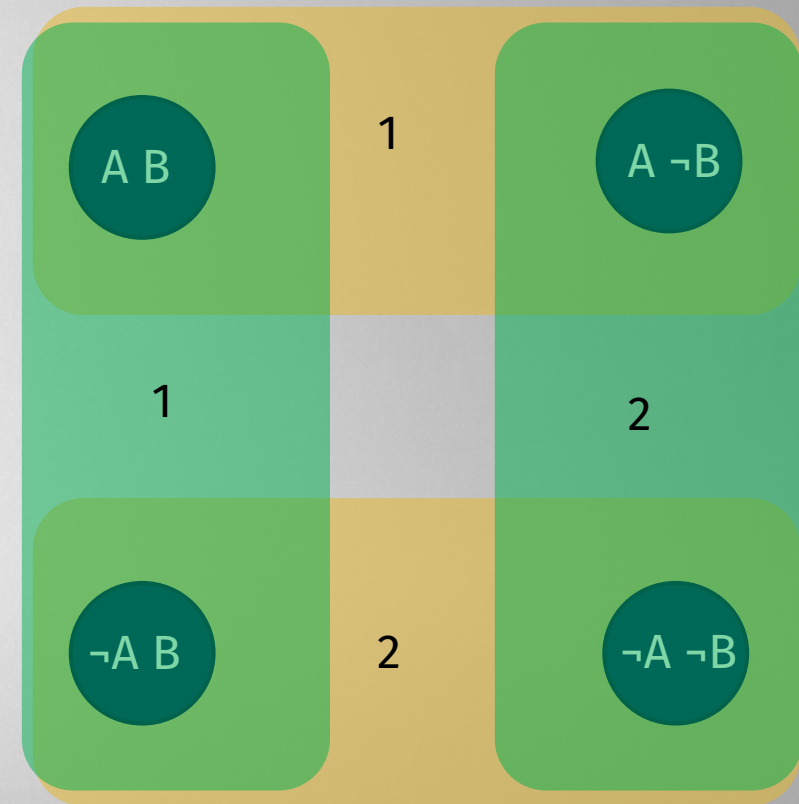
Simplifying the puzzle: Only two professors.

- ▶ Let's represent Prof. B's knowledge, in yellow:
 - ▶ Prof. B knows whether Prof. A has a mistake in his published work or not. Prof. B can distinguish between these two events.
 - ▶ But Prof. B cannot distinguish whether she has a mistake or not in her own work.
 - ▶ The yellow rectangles represent this, the states which Prof. B can distinguish.
 - ▶ As you can see, the states AB and $A\neg B$ form a group, and the states $\neg AB$ and $\neg A\neg B$ form another.
 - ▶ At the true state, $\neg A\neg B$, Prof. B's information cannot reveal the true state.



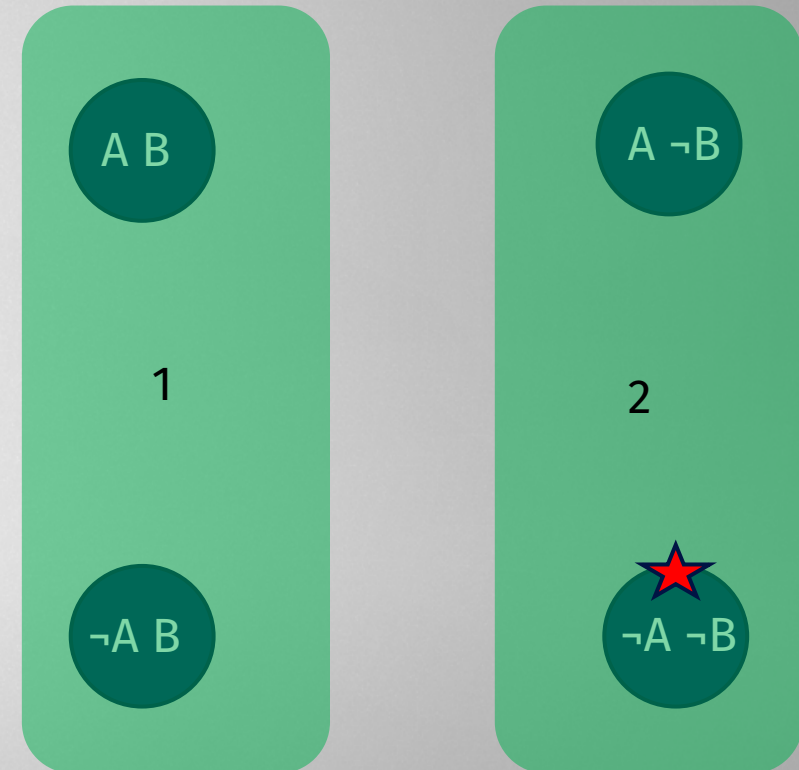
Simplifying the puzzle: Only two professors.

- ▶ Here I am representing the information that both Prof. A and Prof. B possess, at the same time.
- ▶ Their information by itself doesn't reveal the true state, $\neg A \neg B$, for neither of them.
- ▶ The numbers represent the, information groups for each of the professors.
- ▶ At the true state, $\neg A \neg B$, Prof. A is in the green group 2 and Prof. B is in the yellow group 2.
- ▶ After the announcement of Prof. X that "at least one of you has mistakes in your published work", something changes...
- ▶ The professors meet next week for the weekly luncheon... nobody says anything.
- ▶ At the next luncheon, the two of them resign!



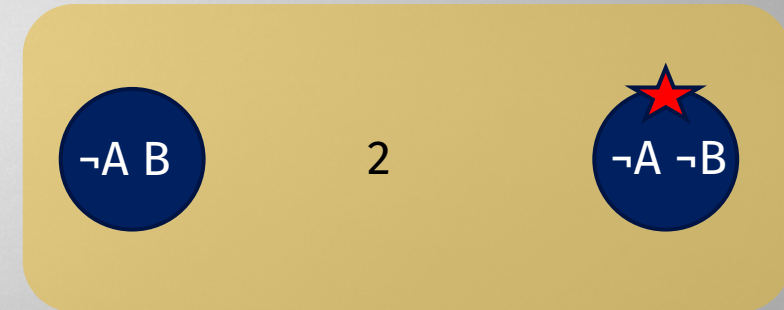
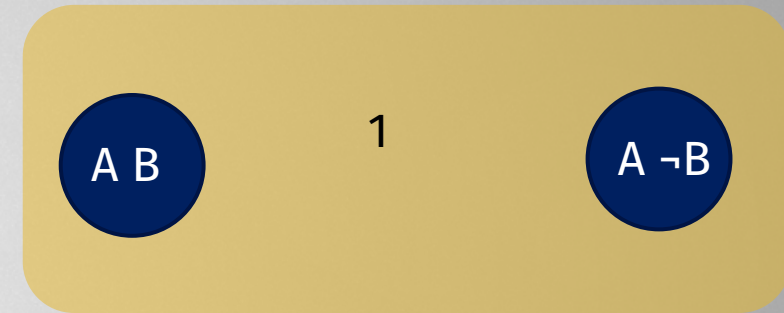
Simplifying the puzzle: Only two professors.

- ▶ After the announcement, and then meeting for the weekly luncheon, Prof. A, which has the information in green group 2, **can no longer think that $A \neg B$ is the true state.**
- ▶ He reasons as follows: If $A \neg B$ were the true state, then Prof. B, after the announcement, would have known that the only one who had mistakes in her work was her. **Since she remained quiet, then Prof. B must know that my work contains a mistake.** Since I also know that Prof. B's work contains a mistake, the true state of affairs is $\neg A \neg B$.
- ▶ Notice here the use of a deeper level of knowledge: Prof. A knows that Prof. B knows that Prof. A's work has a mistake. Therefore, he correctly discovers the true state.



Simplifying the puzzle: Only two professors.

- ▶ After the announcement, and then meeting for the weekly luncheon, Prof. B, which has the information in yellow group 2, **can no longer think that $A \neg B$ is the true state.**
- ▶ She reasons as follows: If $A \neg B$ were the true state, then Prof. A, after the announcement, would have known that the only one who had mistakes in his work was him. **Since he remained quiet, then Prof. A must know that my work contains a mistake.** Since I also know that Prof. A's work contains a mistake, the true state of affairs is $\neg A \neg B$.
- ▶ Notice here the use of a deeper level of **knowledge**: Prof. B knows that Prof. A knows that Prof. B's work has a mistake. Therefore, she correctly discovers the true state.

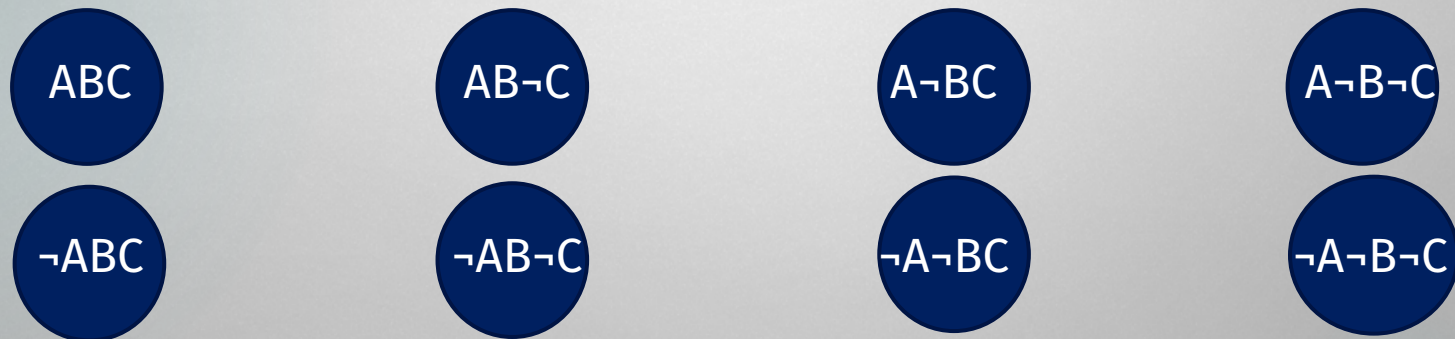


Simplifying the puzzle: Only two professors.

- ▶ What changed after the announcement?
 - ▶ Before Prof. X announcement. It was known by both Professors that “at least one of them had a mistake”. Prof. A knew Prof. B had a mistake and vice versa.
 - ▶ However, this fact was **not commonly known**.
 - ▶ After the **public announcement**, then it is commonly known that “at least one of them had a mistake”.
 - ▶ After their first meeting, then each professor can make the statement “I know that the other one knows that I have a mistake”, which previously they couldn’t do.
 - ▶ After the first meeting, Prof. A expected Prof. B to resign because he knows she has a mistake in her published work. Because she didn’t resign, then Prof. A knows it is because Prof. B is aware of an error in Prof. A’s work. Thus, Prof. A will know about his error and must resign on the next meeting. Similarly for Prof. B.

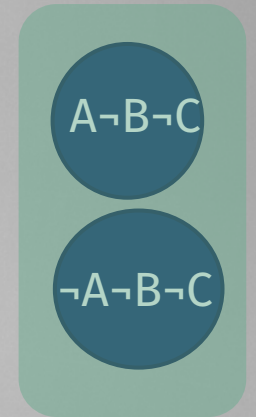
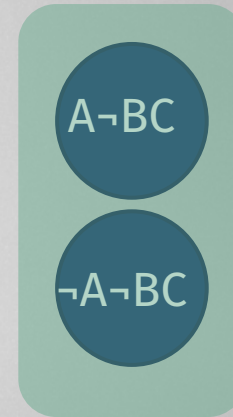
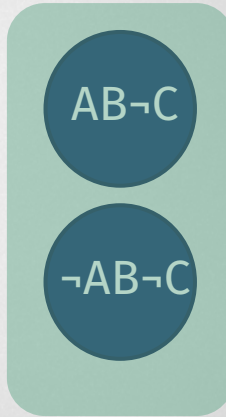
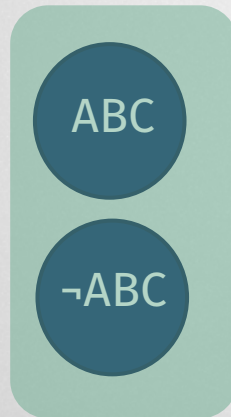
Generalizing a little: three professors.

- ▶ Let's do this problem with three Professors: Prof. A, Prof. B, and Prof. C.
- ▶ For simplicity, let's assume that all of them are males, so I can keep using the pronoun "he".
- ▶ As in the description, all of them have mistakes in their published works.
- ▶ This fact is known to each one of them, but they remain ignorant about mistakes in their own work.
- ▶ Again, let's denote the possible states in this situation: A denotes the event that Prof. A has no mistake in his work and $\neg A$ that he has, similarly for the others. Then we have 8 possible configurations.



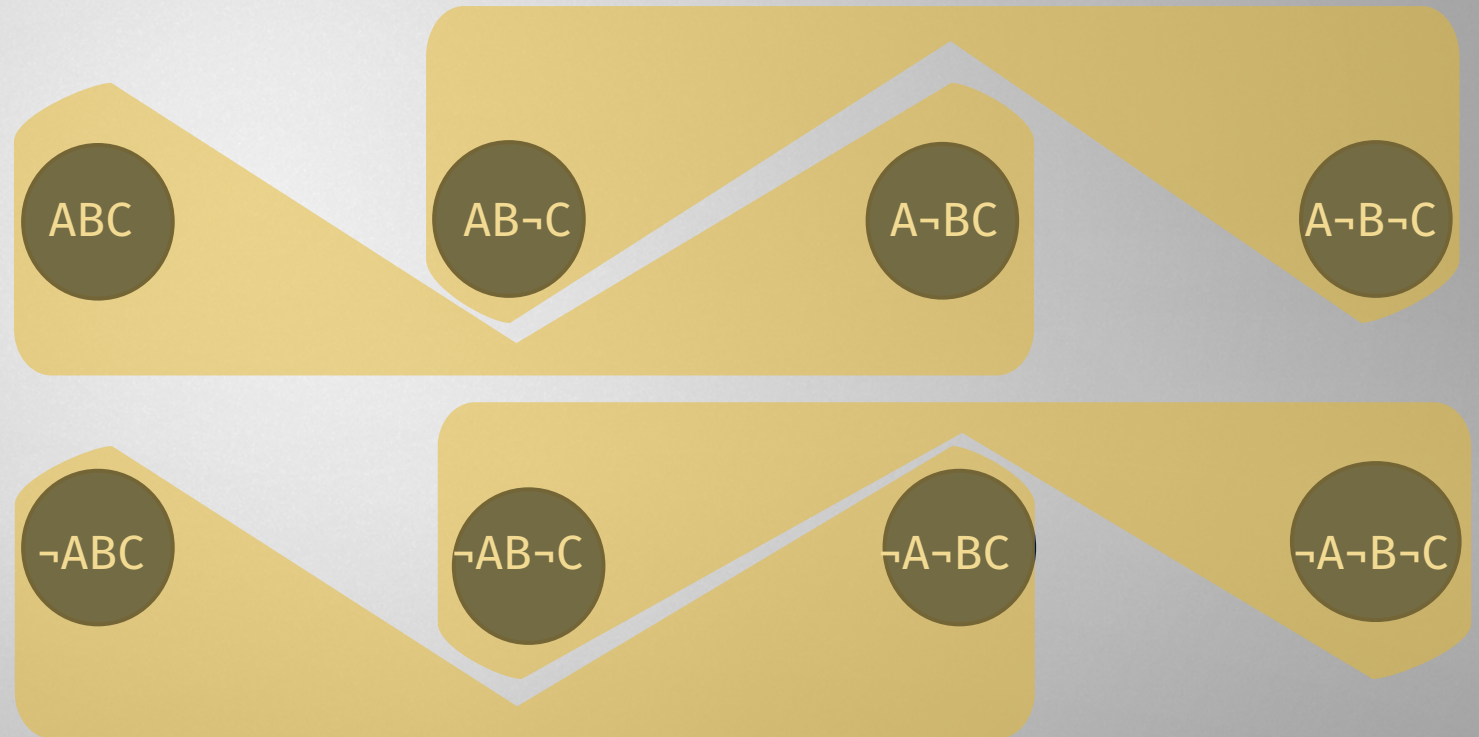
Generalizing a little: three professors.

- ▶ Prof. A's information.
- ▶ Again, we see that at the true state $\neg A \neg B \neg C$, Prof. A can't distinguish the true state from $A \neg B \neg C$.



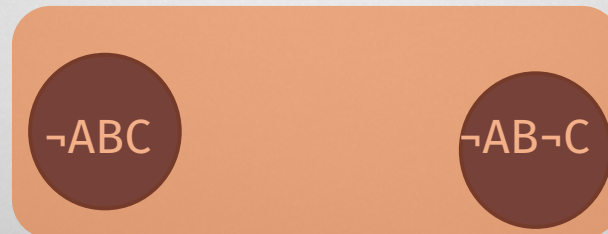
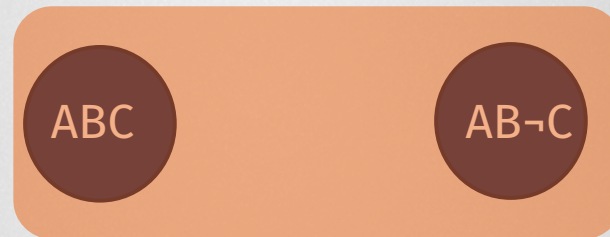
Generalizing a little: three professors.

- ▶ Prof. B's information.
- ▶ Again, we see that at the true state $\neg A \neg B \neg C$, Prof. B can't distinguish the true state from $\neg AB \neg C$.



Generalizing a little: three professors.

- ▶ Prof. C's information.
- ▶ Again, we see that at the true state $\neg A \neg B \neg C$, Prof. B can't distinguish the true state from $\neg A \neg B C$.

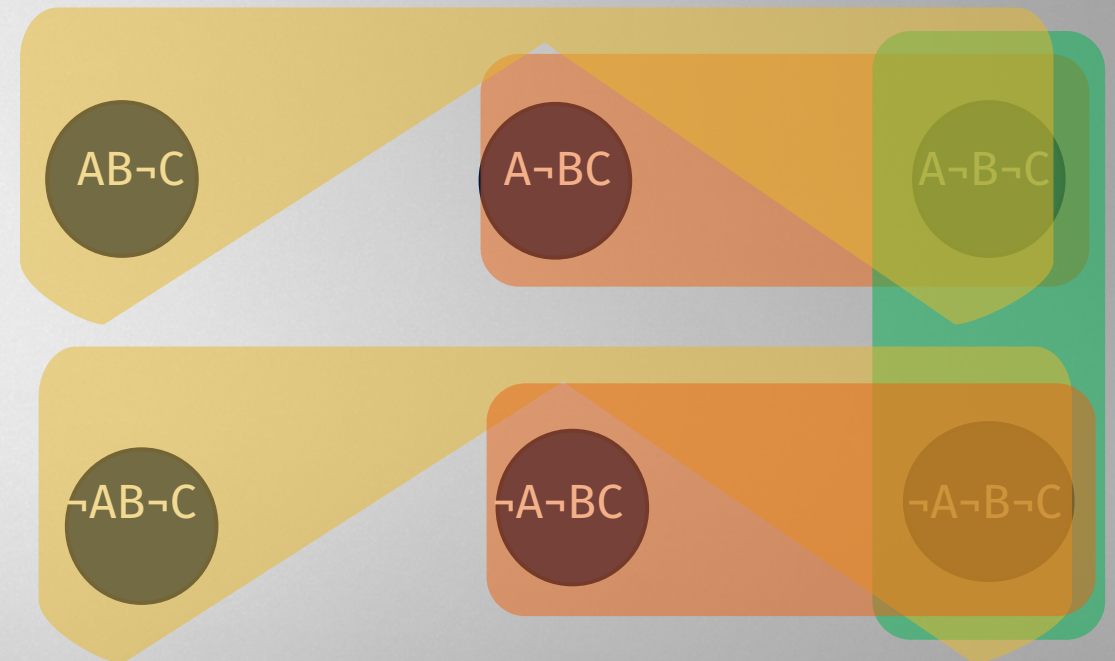


Generalizing a little: three professors.

- ▶ Let's look at what happens here after the announcement, during the first meeting, from the perspective of Prof. A.
- ▶ Prof. A knows errors for both Prof. B and Prof. C.
- ▶ Prof. A knows Prof. B knows about Prof. C's error, and Prof. A knows Prof. C knows about Prof. B's error.
- ▶ Thus Prof. A knows that no one will resign in the first meeting.
- ▶ However, once Prof. C (or Prof. B) doesn't resign, Prof. A knows that Prof. B knows that Prof. C knows about some error.
- ▶ So unless Prof. B knows about two errors for Profs. A and B, then he must resign at the next meeting.

ABC

$\neg ABC$

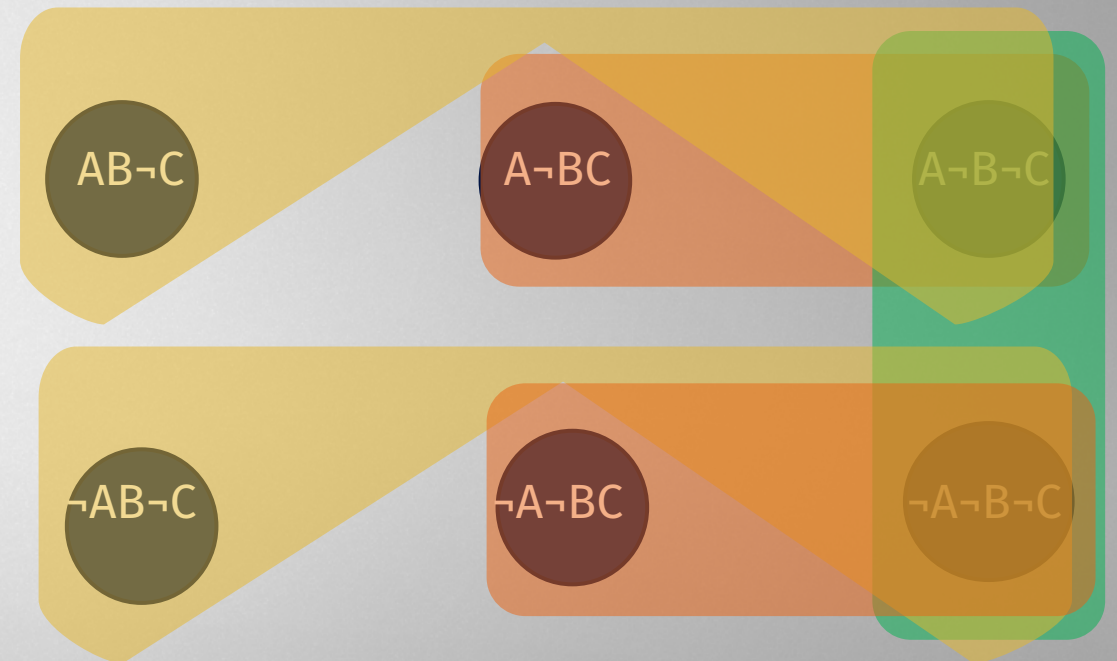


Generalizing a little: three professors.

- ▶ Let's look at what happens here after the announcement, during the second meeting, from the perspective of Prof. A.
- ▶ When Prof. B (or Prof. C) doesn't resign in the second meeting, then Prof. A knows that Prof. B. knows about two errors. And since Prof. B wouldn't know about an error about himself, Prof. B must know and error about Prof. A.
- ▶ Thus, in the third meeting, Prof. A must resign.
- ▶ A similar argument can be made for each of the professors.

ABC

$\neg ABC$



What about the original case, for 17 professors?

- ▶ I am not going to draw the state configuration for the original puzzle, since it involves 17^2 states.
- ▶ However, we can use the intuition from the previous case to give an answer.
- ▶ A professor knows exactly about 16 mistakes (other than his own).
- ▶ After the announcement, everybody still knows about the 16 mistakes other than his own.
- ▶ If no one has resigned before the 16th meeting after Prof. X's statement, then a professor knows all members of the department know at least of 16 errors which are not their own. So the professor now knows there must be at least 17 errors that everyone, except the error maker, knows about. Since he knows only about 16 errors, one of the errors must be his own and he must resign.
- ▶ Therefore, everybody resign at the 16th meeting.
- ▶ To sum up, Prof. X announcement made **common knowledge** the fact that "at least one of you has a mistake.."
- ▶ This allowed each professor to make a chain of reasoning of the type "I know that B knows that C knows that ..."
- ▶ Which ultimately allowed them to correctly infer that everyone had mistakes in their own work.
- ▶ So the public announcement of Prof. X together with the weekly meetings after the fact, imparted extra knowledge to the Professors.

Why is common knowledge relevant?

- ▶ After the detour with the puzzle, let's go back to the analysis of strategic behavior.
- ▶ Let's make the following definition. A strategy s is a **best response** to a particular profile of strategies of my opponents if s maximizes my payoff when my opponents play that particular profile.
- ▶ Thus rational agents choose best responses against their opponents behavior.
- ▶ Rational players can only choose a strategy if it is a best response against the strategies of their opponents.
- ▶ Think about that a best response is like if you were justifying your actions to your boss. If your boss asks, "Why did you choose this course of action?" you would ideally justify it by saying that it was the best that you could possibly do.
- ▶ Now recall: A strategy for a player is strictly dominated if there is another strategy which achieves a higher payoff regardless of what other players do.
- ▶ We have the following equivalence: **A strategy is a best response if and only if it is not dominated by another strategy.**

Common knowledge of rationality

- ▶ If all players are rational
 - ▶ They choose to play best responses or undominated strategies.
- ▶ If all players know that all players are rational
 - ▶ Players believe that their opponents will play best responses or undominated strategies.
 - ▶ They will choose best responses to best responses (the second step in elimination of dominated strategies).
- ▶ If all players know that all players know that all players are rational
 - ▶ Players believe that their opponents will play best responses to best responses.
 - ▶ They will choose best responses to best responses to best responses (third step in elimination).
- ▶ There is common knowledge of rationality.
 - ▶ Players will choose the strategies that survive all steps of the iterated deletion of dominated strategies.

Another example

	A	B	A	C	D
W	5,6	4,4		6,5	12,2
X	3,7	8,7		6,8	10,6
Y	2,10	7,11		4,6	13,5
Z	6,4	5,9		4,10	10,9

- D is strictly dominated by C
- After D is eliminated Y is strictly dominated by X
- After Y is eliminated, B is strictly dominated by C
- After this, no other strategy for any player strictly dominates another one.
- The strategies that remain are “rational”. Each one of them is undominated or a best response to another strategy.
- For example: playing W can be justified by 1 using the following argument:
 - 1 believes that 2 will choose C
 - 1 believes that 2 believes that 1 will chose Z.
 - 1 believes that 2 believes that 1 believes that 2 will choose A
 - 1 believes that 2 believes that 1 believes that 2 believes that 1 will choose W
- To get a stronger prediction in this game, we will need a stronger solution concept.